Air Drawing System Using Hand Gesture Recognition

1. Introduction

The Air Drawing System is a real-time computer vision project that allows users to draw on a virtual canvas using hand gestures in front of a webcam. Instead of using a physical mouse, keyboard, or touchscreen, users can control the drawing process by moving their fingers in the air.

The system uses Python, OpenCV, and MediaPipe to detect hand movements and recognize specific finger gestures. It provides an interactive and touchless drawing experience.

2. Objectives

The main objectives of this project are:

- To develop a real-time hand tracking system using a webcam.
- To detect and track hand landmarks using MediaPipe.
- To recognize finger gestures for drawing and controlling the canvas.
- To implement different drawing colours and an eraser.
- To provide an interactive toolbar for controlling the application.
- To save the completed drawings as image files.

3. Technologies Used

Technology	Purpose
Python	Main programming language
OpenCV	Webcam access, image processing, and drawing
MediaPipe	Hand detection and landmark tracking
NumPy	Canvas creation and image operations

4. System Working Principle

The system works by capturing live video through a webcam and processing each frame using OpenCV.

MediaPipe detects the user's hand and identifies 21 hand landmarks. These landmarks are used to determine the positions of the fingers and recognize hand gestures.

The system uses the following gestures:

- **Index finger only:** Draw on the virtual canvas.
- **Index and middle fingers:** Move the pointer without drawing.
- **Other gestures:** Stop drawing.

When the index finger is detected in drawing mode, the system tracks its position and connects consecutive positions using lines. This creates a continuous drawing effect.

5. Main Features

5.1 Hand Tracking

MediaPipe tracks 21 landmarks on the hand, allowing the system to identify finger positions and movements in real time.

5.2 Air Drawing

Users can draw on the virtual canvas by moving their index finger in front of the webcam.

5.3 Multiple Colours

The system provides four drawing colours:

- Red
- Purple
- Blue
- Green

Users can select colours using the on-screen toolbar.

5.4 Eraser

The eraser feature allows users to remove unwanted parts of their drawings.

5.5 Clear Canvas

The Clear button removes all drawings from the virtual canvas.

5.6 Save Drawing

Users can save their completed drawings as PNG image files. The saved image has a white background.

6. Software Implementation

The project was developed using Python and the MediaPipe Tasks API for hand landmark detection.

The main implementation steps are:

1. Capture video frames using the computer's webcam.
2. Flip the frames horizontally to provide a mirror-like experience.
3. Convert the frames from BGR to RGB format.
4. Detect the hand using MediaPipe.
5. Extract the 21 hand landmarks.
6. Identify the index and middle finger positions.
7. Recognize the drawing and pointer gestures.
8. Draw lines on the virtual canvas based on finger movement.
9. Display the canvas together with the live webcam feed.
10. Allow users to control the application through the on-screen toolbar.

7. Applications

This project demonstrates the potential of gesture-based human-computer interaction. Possible applications include:

- Touchless computer interfaces.
- Interactive smart classrooms.
- Digital presentations.
- Gesture-controlled applications.
- Assistive technologies.
- Gaming and virtual reality systems.

8. Future Improvements

The system can be further improved by introducing:

- More hand gestures for additional controls.
- Adjustable brush thickness.
- Multiple drawing shapes and tools.

- A more advanced graphical user interface.
- Support for drawing on presentation screens.
- Improved gesture recognition and tracking accuracy.

9. Conclusion

The Air Drawing System successfully demonstrates how computer vision and hand gesture recognition can be used to create a touchless drawing application. By combining Python, OpenCV, and MediaPipe, the project provides real-time hand tracking and interactive drawing features.

This project helped develop practical knowledge of computer vision, image processing, hand landmark detection, and gesture-based human-computer interaction.